

Java SE 21 Developer Professional

Duration: -40 hours

1. Introduction to Java

- Course Overview
- Java Features and Object-Oriented Concepts
- Defining, Compiling, and Executing a Java Program
- JDK Installation and Verification
- Creating and Running Java Applications
- **Skill Check (80% pass required)**

2. Primitive Types, Operators, and Flow Control

- Primitive Data Types and Restrictions
- Java Operators: Assignment, Arithmetic, Type Casting
- Binary Numbers and Bitwise Operators
- Conditional, Relational, and Ternary Operators
- If-Else and Switch Statements
- JShell Overview
- Hands-on Practices
- **Skill Check**

3. Text, Date, Time, and Numeric Objects

- String Handling and Mutable Text Objects
- Wrapper Classes and BigDecimal
- Date and Time API
- Localization and Formatting
- Numeric, Date, and Message Formatting
- Hands-on Practices
- **Skill Check**

4. Classes and Objects

- Java Classes and Objects (Parts 1–4)
- Creating and Enhancing Classes
- Java Documentation (Javadoc)
- **Skill Check**

5. Improved Class Design

- Constructor Overloading
- Encapsulation and Immutability
- Enums
- Memory Allocation and Cleanup
- Hands-on Practices
- **Skill Check**

6. Inheritance and Records

- Inheritance and Code Reuse
- Reference Type Casting and instanceof
- Polymorphism and Method Overriding
- Abstract, Final, Sealed Classes
- Records and Pattern Matching
- Factory Methods
- Extensive Hands-on Labs
- **Skill Check**

7. Interfaces and Generics

- Interfaces and Functional Interfaces
- Generics and Type Safety

- Common Java Interfaces
- Composition Design Pattern
- Hands-on Practices
- **Skill Check**

8. Arrays and Loops

- One-Dimensional and Multidimensional Arrays
- Looping Constructs
- Nested Loops, Break, and Continue
- Hands-on Practice
- **Skill Check**

9. Collections Framework

- Collection Interfaces and Implementations
- List, Set, Map, Deque
- Iteration and Sequenced Collections
- Concurrent Access and Thread Safety
- Hands-on Practices
- **Skill Check**

10. Nested Classes and Lambda Expressions

- Inner and Nested Classes
- Anonymous Classes
- Lambda Expressions
- Method References
- Hands-on Practices
- **Skill Check**

11. Java Streams API

- Stream Pipelines and Operations
- Functional Interfaces
- Aggregation and Collectors
- Parallel Streams and Spliterator
- Hands-on Practices
- **Skill Check**

12. Exception Handling, Logging, and Debugging

- Java Logging
- Exception Handling (Advanced)
- Debugging Techniques
- Hands-on Practices
- **Skill Check**

13. Java IO API

- File and Stream Handling
- Serialization and Deserialization
- NIO Path API
- ZIP, HTTP, and Advanced IO
- Hands-on Practices
- **Skill Check**

14. Java Concurrency and Multithreading

- Threads and Executors
- Synchronization and Locks
- Atomic Operations
- Virtual Threads

- Hands-on Practices
- **Skill Check**

15. Modules and Deployment

- Java Module System
- Modular Application Development
- Packaging and Deployment
- Hands-on Practices
- **Skill Check**

16. Annotations

- Built-in and Custom Annotations

17. Java Database Connectivity (JDBC)

- JDBC Architecture
- Database Connectivity and Transactions

18. Java Security

- Java Security Fundamentals
- Secure Application Development

19. Advanced Generics

- Type Erasure
- Advanced Generic Design

20. Java Applications on Oracle Cloud

- Cloud-Native Java Design
- Deploying Java SE Applications

21. Miscellaneous Java Topics

- Design Patterns
- File Watch Service
- Advanced Java IO Concepts

Learning Outcomes – Java SE Comprehensive Training

1. Understand Core Java Concepts

- Explain Java features, JVM architecture, and object-oriented programming principles.
- Install, configure, compile, and execute Java applications using the JDK.

2. Work with Data Types and Control Flow

- Declare and manipulate primitive data types.
- Apply operators, conditional statements, loops, and switch constructs to control program flow.

3. Handle Text, Numbers, and Date/Time APIs

- Perform advanced string manipulation using String, StringBuilder, and StringBuffer.
- Use wrapper classes, BigDecimal, and Java Date/Time APIs.
- Apply localization and formatting for numbers, dates, and messages.

4. Design and Implement Classes

- Create well-structured classes and objects.
- Apply encapsulation, immutability, constructors, and method overloading.
- Use enums and follow Java naming and documentation standards.

5. Apply Inheritance and Polymorphism

- Extend classes and reuse code using inheritance.
- Override methods and implement polymorphic behavior.
- Use abstract, final, sealed classes, and records effectively.

6. Develop with Interfaces and Generics

- Design and implement interfaces and functional interfaces.
- Apply generics to achieve type safety and reusable code.
- Use composition patterns for scalable design.

7. Work with Arrays and Collections

- Create and manipulate arrays and multidimensional data structures.
- Use Java Collections Framework (List, Set, Map, Queue) efficiently.
- Implement sorting, searching, and concurrent-safe collections.

8. Use Functional Programming Constructs

- Implement lambda expressions and method references.
- Process data using the Java Streams API.
- Perform filtering, mapping, grouping, and aggregation operations.

9. Handle Exceptions, Logging, and Debugging

- Implement robust exception handling strategies.
- Use Java logging frameworks for application monitoring.
- Debug Java applications using industry-standard techniques.

10. Perform File and I/O Operations

- Read and write files using Java IO and NIO APIs.
- Implement serialization and deserialization.
- Work with file systems, ZIP archives, and external resources.

11. Develop Concurrent and Multithreaded Applications

- Create and manage threads and executor services.
- Ensure thread safety using synchronization, locks, and atomic operations.
- Apply virtual threads and concurrency best practices.

12. Build Modular and Deployable Applications

- Create modular Java applications using the Java Module System.
- Package, deploy, and migrate applications effectively.

13. Use Annotations, JDBC, and Security APIs

- Create and use Java annotations.
- Connect Java applications to databases using JDBC.
- Apply Java security principles for secure application development.

14. Apply Advanced Java Concepts

- Use advanced generics and design patterns.
- Develop cloud-ready Java applications and deploy them on Oracle Cloud.

15. Build Real-World Java Applications

- Design and implement a complete Java application using industry best practices.
- Apply object-oriented, functional, and concurrent programming concepts in real-world scenarios.